Unit 32: Exponential Functions — Full Solutions

Warm-up

1. $\dfrac{d}{dx}[e^x] = e^x$

2. $\dfrac{d}{dx}[e^{2x}]$

Chain rule: $u = 2x$, $u' = 2$.

$$\frac{d}{dx}[e^{2x}] = 2e^{2x}$$

3. $\dfrac{d}{dx}[3e^x] = 3e^x$

4. Solve $e^x = 5$.

$$x = \ln 5$$

5. $\displaystyle\int e^x\,dx = e^x + C$

6. $\dfrac{d}{dx}[e^x + x^2] = e^x + 2x$

Standard

7. $\dfrac{d}{dx}[e^{x^2}]$

Chain rule: $u = x^2$, $u' = 2x$.

$$\frac{d}{dx}[e^{x^2}] = 2xe^{x^2}$$

8. $\dfrac{d}{dx}[e^{3x+1}]$

Chain rule: $u = 3x+1$, $u' = 3$.

$$\frac{d}{dx}[e^{3x+1}] = 3e^{3x+1}$$

9. $\dfrac{d}{dx}[xe^x]$

Product rule: $f = x$ $(f' = 1)$, $g = e^x$ $(g' = e^x)$.

$$\frac{d}{dx}[xe^x] = 1 \cdot e^x + x \cdot e^x = e^x(1+x)$$

10. $\dfrac{d}{dx}\left[\dfrac{e^x}{x}\right]$

Quotient rule: $f = e^x$ ($f' = e^x$), $g = x$ ($g' = 1$).

$$\frac{d}{dx}\left[\frac{e^x}{x}\right] = \frac{e^x \cdot x - e^x \cdot 1}{x^2} = \frac{e^x(x-1)}{x^2}$$

11. Solve $e^{2x-1} = 7$.

$$2x - 1 = \ln 7$$

$$2x = \ln 7 + 1$$

$$x = \frac{\ln 7 + 1}{2}$$

12. $\displaystyle\int 3e^{2x}\,dx$

Let $u = 2x$, $du = 2\,dx \Rightarrow dx = \dfrac{du}{2}$.

$$3\int e^u \cdot \frac{du}{2} = \frac{3}{2}e^u + C = \frac{3}{2}e^{2x} + C$$

13. $\displaystyle\int_0^1 e^x\,dx$

$$\Big[e^x\Big]_0^1 = e^1 - e^0 = e - 1$$

Answer: $e - 1$.

Challenge

14. Solve $e^{3x-6} = 9$.

$$3x - 6 = \ln 9$$

$$3x = \ln 9 + 6$$

$$x = \frac{\ln 9 + 6}{3}$$

(Since $9=3^2$, this can also be written as $x=\dfrac{2\ln 3+6}{3}=\dfrac{2\ln 3}{3}+2$.)

15. $y=(9x^2-6x+2)e^{3x}$. Find $\dfrac{dy}{dx}$.

Product rule: $f=9x^2-6x+2$ ($f'=18x-6$), $g=e^{3x}$ ($g'=3e^{3x}$).

$$\frac{dy}{dx}=(18x-6)e^{3x}+(9x^2-6x+2)(3e^{3x})$$

Factor out e^{3x}:

$$=e^{3x}\Big[(18x-6)+3(9x^2-6x+2)\Big]=e^{3x}[18x-6+27x^2-18x+6]$$

Notice the $18x$ and $-18x$ cancel, and $-6+6=0$:

$$=e^{3x}(27x^2)=27x^2e^{3x}$$

$\dfrac{dy}{dx}=27x^2e^{3x}$.

16. $\dfrac{d}{dx}[e^{\sin x}]$

Chain rule: $u=\sin x$, $u'=\cos x$.

$$\frac{d}{dx}[e^{\sin x}]=e^{\sin x}\cdot\cos x$$

17. $\displaystyle\int 2xe^{x^2}\,dx$

Let $u=x^2$, $du=2x\,dx$.

$$\int e^u\,du=e^u+C=e^{x^2}+C$$

18. Solve $2e^x-3=7$.

Isolate the exponential term first:

$$2e^x=10$$

$$e^x=5$$

$$x=\ln 5$$

Applied

19. $A(t) = 1000e^{0.06t} = 2000$ (doubled).

$$e^{0.06t} = 2$$

$$0.06t = \ln 2$$

$$t = \frac{\ln 2}{0.06} \approx \frac{0.6931}{0.06} \approx 11.55 \text{ years}$$

20. $P(t) = 200e^{0.03t}$

$$\frac{dP}{dt} = 200(0.03)e^{0.03t} = 6e^{0.03t}$$

At $t = 10$:

$$\left.\frac{dP}{dt}\right|_{t=10} = 6e^{0.3} \approx 6(1.3499) \approx 8.10$$

Answer: $\frac{dP}{dt} = 6e^{0.03t}$, and at $t = 10$, the growth rate is ≈ 8.10 people per year.

21. $C(t) = 50e^{-0.5t}$

$$\frac{dC}{dt} = 50(-0.5)e^{-0.5t} = -25e^{-0.5t}$$

At $t = 2$:

$$\left.\frac{dC}{dt}\right|_{t=2} = -25e^{-1} \approx -25(0.3679) \approx -9.20$$

Answer: $\frac{dC}{dt} = -25e^{-0.5t}$, and at $t = 2$, the concentration is decreasing at a rate of approximately 9.20 mg per unit time.

22. $D(t) = 80e^{-0.1t}$, from $t = 0$ to $t = 10$.

Let $u = -0.1t$, $du = -0.1\,dt \Rightarrow dt = -10\,du$.

$$\int 80e^{-0.1t}\,dt = 80\int e^u(-10)\,du = -800\int e^u\,du = -800e^u + C = -800e^{-0.1t} + C$$

Evaluate from 0 to 10:

$$\Big[-800e^{-0.1t}\Big]_0^{10} = (-800e^{-1}) - (-800e^{0}) = -800e^{-1} + 800 = 800(1-e^{-1})$$

$$\approx 800(1-0.3679) \approx 800(0.6321) \approx 505.7$$

Answer: total accumulated area $= 800(1-e^{-1}) \approx 505.7$.